



Review of correlations for subcooled flow boiling heat transfer and assessment of their applicability to water

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ABSTRACT

The subcooled flow boiling heat transfer of water has many applications, such as cooling devices of nuclear power systems and electronics, and the applicability of correlations of subcooled flow boiling heat transfer to water needs to be assessed. A number of correlations for subcooled flow boiling heat transfer were proposed. However, their prediction accuracy for water was not systematically investigated. This paper presents a comprehensive review of correlations of subcooled flow boiling heat transfer. A large database of subcooled flow boiling heat transfer of water containing 1184 data points is compiled from 14 published articles, based on which 21 correlations of subcooled flow boiling heat transfer are assessed, and their prediction accuracy is analyzed for different channel sizes, channel shapes, and flow directions. The results are helpful for understanding the prediction methods of subcooled flow boiling heat transfer and choosing a proper correlation for a given application.

1. Introduction

Subcooled flow boiling heat transfer of water in channels is widely used in plasma facing components of fusion nuclear systems, thermal power plants, as well as electronic cooling systems [1–5]. The calculation of subcooled flow boiling heat transfer of water is important for designing such facilities. A number of correlations for subcooled flow boiling heat transfer were proposed, of which some are also employed for more sophisticated geometries, such as hypervapotron [6] and swirl tubes [7].

There are a variety of factors that affect subcooled flow boiling heat transfer, such as fluid, mass flux, heat flux, subcooling, pressure, and channel geometry, which makes it a challenge to predict subcooled flow boiling heat transfer. In the past 70 years, many investigations were conducted to determine subcooled flow boiling heat transfer. Due to the complexity of this issue, the experimental study has been the main research approach, and models for calculating subcooled flow boiling heat transfer are mostly experiment-based correlations, among which most were based on experimental data with water.

Such intensive investigations have led to a better understanding of subcooled flow boiling heat transfer characteristics and mechanisms. Meanwhile, over 20 correlations of subcooled flow boiling heat transfer have been proposed [2,3,5,8–26]. A large number of correlations provide the engineers and designers more choices in engineering practice. On the other hand, they are confronted with considerable confusion in

choosing a suitable correlation for a given application since it is hard for them to know which correlation may yield a better calculation result.

Neither the systematic review on the correlations of subcooled flow boiling heat transfer nor the assessment of their applicability to water with a multiple-source database has been found. The goal of the present work is to review the correlations of subcooled flow boiling heat transfer and to evaluate their applicability to water based on a large experimental database compiled from 14 published articles, which contains 1184 data points of subcooled flow boiling heat transfer of water in various channel sizes and both vertical and horizontal flow directions over a wide range of parameters. Twenty-two existing correlations of subcooled flow boiling heat transfer are reviewed, among which 21 are assessed and analyzed using the database. The results provide a strong support for understanding prediction methods of subcooled flow boiling heat transfer and choosing a proper correlation for a given application.

2. Overview of correlations for subcooled flow boiling heat transfer

This section presents a systematic review of correlations for subcooled flow boiling heat transfer. These correlations can be classified into five categories: enhancement-factor, superposition, asymptotic, $q \sim \Delta T_{sat}^n$, and flow pattern based types. The wall superheat ΔT_{sat} is

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Nomenclature		X_{tt}	Martinelli parameter for turbulent liquid/turbulent gas
		<i>Greek symbols</i>	
Bo	Boiling number	β	Volumetric thermal expansion coefficient (1/K)
c_p	Specific heat at constant pressure (J/kg K)	λ	Thermal conductivity (W/m K)
D	Diameter, hydraulic diameter (m)	μ	Dynamic viscosity (kg/s m)
F	Reynolds number factor	ρ	Density (kg/m ³)
Fr	Froude number	σ	Surface tension (N/m)
f	Moody friction factor	ψ	Enhancement factor
G	Mass flux (kg/m ² s)	<i>Subscripts</i>	
H	Channel height (m)	b	Bulk temperature
h	Heat transfer coefficient (W/m ² K)	cp	Constant property
h_{ig}	Latent heat of vaporization (J/kg)	$crit$	Critical point
Ja	Jacob number	cv	Convection
Ja^*	Modified Jacob number	exp	Experimental
L	Channel length (m)	f	Film
M	Molecular mass (kg/kmol)	fc	Forced convection
Nu	Nusselt number	g	Gas phase
p	Pressure (Pa)	in	Inlet
P_R	Reduced pressure, $P_R = p/p_{crit}$	l	Liquid phase
Pr	Prandtl number	lo	Liquid only, assuming all fluid as liquid
q	Heat flux from channel wall to fluid (W/m ²)	loc	Local
Re	Reynolds number	nb	Nucleate boiling
S	Suppression factor	nc	Natural convection
T	Temperature (K)	$pred$	Predicted
ΔT_{sat}	Wall superheat (K)	sat	Saturated
$\Delta T_{sat,ONB}$	Wall superheat at onset of nucleate boiling (K)	sp	Single-phase
ΔT_{sub}	Subcooling (K)	tp	Two-phase
u	Velocity (m/s)	w	Channel inner wall surface
W	Channel width (m)		
We^*	Modified Weber number		

defined as the difference between the inner wall temperature T_w and the saturated temperature T_{sat}

$$\Delta T_{sat} = T_w - T_{sat} \quad (1)$$

2.1. Enhancement-factor models

For enhancement-factor models, the subcooled flow boiling heat transfer coefficient h_{sp} may be reduced to

$$h_{sp} = \psi h_{sp,l} \quad (2)$$

where ψ is the enhancement factor, which accounts for the enhancement of the boiling to the subcooled flow boiling heat transfer, and $h_{sp,l}$ is the forced convection heat transfer coefficient of the single-phase liquid and can be generalized as

$$h_{sp,l} = C Re_l^m Pr_l^n \frac{\lambda_l}{D} \quad (3)$$

where constants C , m and n need to be determined by curve fitting of experimental data. Substituting Eq. (3) into Eq. (2) yields

$$h_{sp} = Y Re_l^m Pr_l^n \frac{\lambda_l}{D} \quad (4)$$

or

$$Nu_{sp} = h_{sp} \frac{D}{\lambda_l} = Y Re_l^m Pr_l^n \quad (5)$$

where Y is a fluid and flow condition dependent function, which can be determined through curve fitting of experimental data. The liquid Prandtl number Pr_l and the liquid Reynolds number Re_l are defined as

$$Pr_l = \frac{c_{pl} \mu_l}{\lambda_l} \quad (6)$$

$$Re_l = \frac{G_l D}{\mu_l} \quad (7)$$

where G_l is the liquid mass flux.

Since the vapor mass flux is far smaller than the liquid mass flux in subcooled flow boiling, it is reasonable to simplify the calculation by substituting G for G_b , where G is the total mass flux of the two-phase flow. Thus,

Table 1
Values of the constant C_{sf} in the Rohsenow correlation (from Ghiaasiaan [28]).

Fluid-surface combination	C_{sf}	Fluid-surface combination	C_{sf}
Water–nickel	0.006	Benzene–chromium	0.01
Water–platinum	0.013	<i>n</i> -Pentane–chromium	0.015
Water–emery polished copper	0.0128	<i>n</i> -Pentane–emery polished copper	0.0154
Water–brass	0.006	<i>n</i> -Pentane–emery polished nickel	0.0127
Water–ground and polished stainless steel	0.008	Ethyl alcohol–chromium	0.0027
Water–Teflon pitted stainless steel	0.0058	Isopropyl alcohol–copper	0.0025
Water–chemically etched stainless steel	0.0133	35% K ₂ CO ₃ –copper	0.0054
Water–mechanically polished stainless steel	0.0132	50% K ₂ CO ₃ –copper	0.0027
Water–emery polished, paraffin treated copper	0.0147	<i>n</i> -Butyl alcohol–copper	0.0030
CCl ₄ –emery polished copper	0.007		

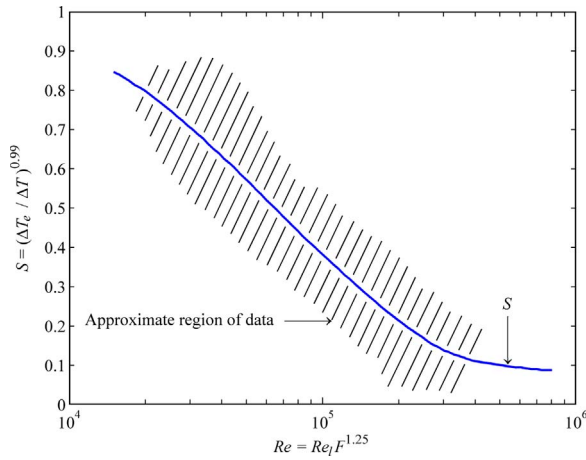


Fig. 1. Suppression factor S provided by Chen [29].

$$Re_l \approx \frac{GD}{\mu_l} = Re_{lo} \quad (8)$$

where Re_{lo} is the liquid only Reynolds number. Liquid only (lo) means assuming that the two-phase flow is the liquid with mass flux equal to that of the two-phase flow.

2.1.1. Shah [5] correlation

Shah compiled a database containing 500 experimental data of subcooled flow boiling heat transfer from 18 independent experimental studies, with fluids including water, R11, R12, R113, ammonia, isopropyl alcohol, n -butyl alcohol, methanol, and aqueous solutions of potassium carbonate in horizontal and vertical pipes and annuli. The range of parameters covered $q = 0.1$ – 22.9 MW/m², $G = 55$ – $24,200$ kg/m²s, $\Delta T_{sub} = 0$ – 153 °C, $p = 0.1$ – 13.6 MPa, $D = 2.4$ – 27.1 mm, $Bo = 0.4 \times 10^{-4}$ – 54×10^{-4} , $Re_l = 1.4 \times 10^3$ – 760×10^3 , and $Pr_l = 0.005$ – 0.708 . With his database he proposed

$$q = h_{tp} \Delta T_{sat} = \psi h_{sp,l} \Delta T_{sat} \quad (9)$$

where the enhancement factor is determined by

$$\psi = \begin{cases} \psi_0 & \text{for low subcooling} \\ \psi_0 + \Delta T_{sub} / \Delta T_{sat} & \text{for high subcooling} \end{cases} \quad (10a)$$

$$\psi_0 = \begin{cases} 230 Bo^{0.5} & \text{for } Bo > 0.3 \times 10^{-4} \\ 1 + 46 Bo^{0.5} & \text{for } Bo < 0.3 \times 10^{-4} \end{cases} \quad (10b)$$

where the boiling number Bo and the subcooling ΔT_{sub} are defined as

$$Bo = \frac{q}{G h_{lg}} \quad (11)$$

$$\Delta T_{sub} = T_{sat} - T_b \quad (12)$$

where T_b is the bulk fluid temperature. The forced convection heat transfer coefficient of single-phase liquid, $h_{sp,l}$ is calculated with the Dittus–Boelter [27] correlation,

$$h_{sp,l} = 0.023 Re_l^{0.8} Pr_l^{0.4} \frac{\lambda_l}{D} \quad (13)$$

2.1.2. Hata and Noda [9] correlation

Hata and Noda experimentally investigated the subcooled boiling of water flowing upward in vertical platinum tubes with inner diameters (IDs) of 3, 6, and 9 mm and length from 32.7 to 100 mm. The ratio of length to ID is between 5.5 and 33.3. The range of experimental parameters covered velocity from 4 to 21 m/s (corresponding to the liquid Reynolds number from 6.8×10^4 to 106.3×10^4), inlet subcooling ($\Delta T_{sub,in} = T_{sat} - T_{b,in}$) from 92.4 to 147.5 °C, and inlet pressure from 0.81 to 1.014 MPa. The subcooled flow boiling heat transfer

coefficient h_{tp} is determined by

$$q = h_{tp} (T_w - T_b) \quad (14)$$

Considering the entrance effects on short tubes, they proposed the following correlation based on their experimental data:

$$h_{tp} = \frac{Nu \lambda_l}{D} = 0.02 Re_l^{0.85} Pr_l^{0.4} \left(\frac{L}{D} \right)^{-0.08} \left(\frac{\mu_{l,b}}{\mu_{l,w}} \right)^{0.14} \frac{\lambda_l}{D} \quad (15)$$

where $\mu_{l,b}$ and $\mu_{l,w}$ are the liquid dynamic viscosities valued at the liquid bulk temperature and the inner wall temperature, respectively, and all fluid properties, except for $\mu_{l,w}$, are valued at the liquid bulk temperature.

2.1.3. Hata and Masuzaki [10] correlation

Hata and Masuzaki conducted an experimental study of subcooled flow boiling of water in a vertical upward SUS304 tube of 6 mm ID and 59.5 mm heated length ($L = 59.5$ mm), with $L/D = 9.92$. The range of the experimental parameters covered velocity from 17.2 to 42.4 m/s (corresponding to the liquid Reynolds number from 5.8×10^5 to 14.3×10^5), inlet subcooling from 80.9 to 147.6 °C, and inlet pressure from 0.81 to 1.18 MPa.

Their study supported the Hata and Noda Eq. (15), except for the fully developed nucleate boiling curves for the flow velocity lower than 21 m/s (corresponding to the liquid Reynolds number lower than 7×10^5) and the nucleate boiling curves in the higher heat flux range for the flow velocity higher than 30 m/s (corresponding to the liquid Reynolds number higher than 10×10^5), where the proposed correlation is

$$q = 51.25 \Delta T_{sat}^3 \quad (16)$$

2.1.4. Baburajan et al. [11] correlation

Baburajan et al. experimentally studied the subcooled flow boiling heat transfer of water in horizontal tubes of 5.5 mm, 7.5 mm and 9.5 mm IDs under low pressure and low flow conditions, with mass fluxes of 450–935 kg/m²s and inlet subcoolings of 29, 50, and 70 °C. They used Eq. (14) to determine the local two-phase heat transfer coefficient h_{tp} and Eq. (2) to determine the enhancement factor ψ , with $h_{sp,l}$ calculated by

$$h_{sp,l} = 0.023 Re_l^{0.8} Pr_l^{0.4} \left(\frac{\mu_{l,b}}{\mu_{l,w}} \right)^{0.262} \frac{\lambda_l}{D} \quad (17)$$

The resultant enhancement factor ψ is of the form

$$\psi = 267 Bo^{0.86} Ja^{*-0.6} Pr_l^{0.23} \quad (18)$$

where the fluid properties are valued at the local bulk fluid temperature, and the modified Jacob number they used is given by

$$Ja^* = \frac{c_{pl} \Delta T_{sub,in}}{h_{lg}} \quad (19)$$

where $\Delta T_{sub,in}$ is the inlet subcooling, $\Delta T_{sub,in} = T_{sat} - T_{l,in}$.

2.1.5. Papell [8] correlation

By using distilled water flowing through a horizontal tube of 7.9 inner diameter and compiling comparable heat transfer data from seven reference sources, Papell [8] obtained a water subcooled flow boiling heat transfer database containing 275 data points. The database covered a broad range of conditions, with pressure from 0.11 to 13.8 MPa, heat flux from 42.5 to 91,580 kW/m², fluid velocity from 0.41 to 62.2 m/s, and subcooling from 3.3 to 186.7 °C. Based on the database, the following correlation was proposed:

$$\frac{Nu_{tp}}{Nu_{sp}} = 90.0 Ja^{*-0.84} \left(\frac{q}{h_{lg} \rho_g u} \right)^{0.7} \left(\frac{\rho_g}{\rho_l} \right)^{0.756} \quad (20)$$

Table 2

Asymptotic type models of subcooled flow boiling heat transfer.

Authors	Correlations	Comments
Hua et al. [2]	$q^{8.5} = [h_{sp,l}(\Delta T_{sat} + \Delta T_{sub})]^{8.5} + \left\{ \lambda_l \mu_l \sqrt{\frac{\sigma}{g(\rho_l - \rho_g)}} \left(\frac{c_{pl} \Delta T_{sat}}{0.046 \lambda_l Pr_l} \right)^{1/0.624} \right\}^{8.5}$ $h_{sp,l} = 2.906 Re_l^{0.36} Pr_l^{0.48} \left(\frac{\mu_{l,b}}{\mu_{l,w}} \right)^{-0.22} \left(\frac{\lambda_l}{D} \right)$	Water flowing in a 24×14 mm horizontal rectangular channel; $u = 0.2\text{--}2.0$ m/s $(Re_l = 9100\text{--}91,000)$, $q = 100\text{--}1100$ kW/m², $p = 100, 200, 300$ kPa, $T_{in} = 75, 85, 95$ °C.
Kutateladze [17]	The wall heat flux q is calculated with Eq. (9), where h_{sp} is given by $h_{ip}^2 = h_{sp,l}^2 + h_{nb}^2$ where $h_{sp,l}$ is calculated with the Dittus–Boelter Eq. (13), and h_{nb} is calculated with $h_{nb} = C(p) q^{0.7}$ where $C(p)$ is an empirical parameter depending on the pressure p (Fig. 2).	Author provided a $p - C(p)$ chart (Fig. 2) based on the subcooled flow boiling heat transfer of water.
Bergles and Rohsenow [18]	$q^2 = q_{sp,l}^2 + q_{nb}^2 \left(1 - \frac{q_{ONB}}{q_{nb}} \right)^2$ where q_{ONB} is the heat flux at onset of nucleate boiling (ONB), $q_{ONB} = 1082 p^{1.156} (1.8 \Delta T_{sat, ONB})^{2.16} / p^{0.0234}$ where p is in bar, and $\Delta T_{sat, ONB}$ is the wall superheat at ONB. $\Delta T_{sat, ONB} = T_{w, ONB} - T_{sat}$ (47b) For q_{nb}, Bergles and Rohsenow provided a chart based on the experimental data, without any mathematical model. It is suggest that a pool boiling correlation may be chosen for calculating q_{nb}, such as the Cooper [32] correlation, Eq. (40).	Water flowing in a horizontal tube with $D = 2.4$ mm.
Liu and Winterton [19]	$q^2 = (F q_{sp,l})^2 + (S q_{nb})^2 = [F h_{sp,l} (\Delta T_{sat} + \Delta T_{sub})]^2 + (S h_{nb} \Delta T_{sat})^2$ $F = [1 + x Pr_l (\rho_l / \rho_g - 1)]^{0.35}$ $S = 1 / (1 + 0.055 F^{0.1} Re_l^{0.16})$ The single-phase forced convection heat transfer coefficient $h_{sp,l}$ is calculated with the Dittus–Boelter correlation, Eq. (13), and the nucleate boiling heat transfer coefficient h_{nb} is calculated with the Cooper [32] correlation, Eq. (40). All fluid properties are valued at the saturation temperature. Since it is reasonable to assume the vapor quality $x = 0$ in subcooled flow boiling, it follows that $F = 1 \text{ (49a),}$ $S = \frac{1}{1 + 0.055 Re_l^{0.16}}$	Based on 4202 data points for saturated boiling and 991 data points for subcooled boiling, collected from 30 different literature sources involving nine different fluids.
Bjorge et al. [20]	$q^2 = q_{sp,l}^2 + q_{nb}^2 \left[1 - \left(\frac{\Delta T_{sat, ONB}}{\Delta T_{sat}} \right)^3 \right]^2$ where $q_{sp,l}$ is determined by Eq. (26), with $h_{sp,l}$ given by	Verified with water data in the range of $D = 10\text{--}13.2$ mm, $q = 150\text{--}2100$ kW/m², $G = 470\text{--}1880$ kg/m²s $(Re_l = 12400\text{--}65700)$,

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Table 2 (continued)

Authors	Correlations	Comments
	$h_{sp,l} = 0.023 Re_f^{0.8} Pr_f^{1/3} \frac{\lambda_f}{D} \quad (51)$ where the subscript f denotes the parameter is valued at the film temperature T_f, $T_f = (T_w + T_b)/2$. The wall superheat at ONB, $\Delta T_{sat,ONB}$, depends on the radius of the point of tangency, r_{tang}, and the radius of the largest cavity, r_{max}, $r_{tang} = \frac{4\sigma T_{sat} \left(\frac{1}{\rho_g} - \frac{1}{\rho_l} \right)}{h_{lg} \Delta T_{sat,ONB}} \quad (52a)$ $\Delta T_{sat,ONB} = \begin{cases} \frac{1}{1-N} \left(\frac{1}{4\Gamma N} - N \Delta T_{sub} \right) & r_{tang} > r_{max} \\ \frac{1}{2\Gamma} [1 + (1 + 4\Gamma \Delta T_{sub})^{1/2}] & r_{tang} < r_{max} \end{cases} \quad (52b)$ $\Gamma = \frac{\lambda_l h_{lg}}{8\sigma T_{sat} h_{sp,l} \left(\frac{1}{\rho_g} - \frac{1}{\rho_l} \right)} \quad (52c),$ $N = \frac{h_{sp,l} r_{max}}{\lambda_l} \quad (52d)$ $q_{nb} = 1.89 \times 10^{-4} \left[\frac{\lambda_l^{1/2} \rho_l^{17/8} c_{pl}^{19/8} h_{lg}^{1/8} \rho_g^{1/8}}{(\rho_l - \rho_g)^{9/8} \sigma^{5/8} T_{sat}^{1/8}} \sqrt{\frac{g(\rho_l - \rho_g)}{\sigma}} \right] \Delta T_{sat}^3 \quad (53)$ The value of $r_{max} = 10^{-6}$ m was used.	$p = 200\text{--}410$ kPa, and $\Delta T_{sub} = 10\text{--}70$ °C.
Ramstorfer et al. [21]	$q^{10} = q_{cv}^{10} + q_{nb}^{10} \quad (54)$ $q_{cv}^{2.5} = q_{fc}^{2.5} + q_{nc}^{2.5} \quad (55a),$ $q_{fc} = [230(Gh_{lg})^{-0.5} h_{sp,l} \Delta T_{sat}]^2 \quad (55b)$ $h_{sp,l} = 0.0447 Re_l^{0.75} Pr_l^{0.4} \frac{\lambda_l}{D} \quad (55c)$ $q_{nc} = \left\{ 0.825 + 0.387 [Gr_l Pr_l f_1(Pr_l)]^{1/6} \right\} \frac{\lambda_l}{L} (T_w - T_b) \quad (55d)$ $Gr_l = \frac{gL^3}{(\mu_l/\rho_l)^2 \beta_l (T_w - T_b)} \quad (55e)$ $q_{nb} = \left(\frac{1}{0.0172} \frac{c_{pl}}{h_{lg}} Pr_l^{-1} \right)^{1/0.9} \left[\frac{\sigma}{g(\rho_l - \rho_g)} \right]^{-1/2} h_{lg} \mu_l \Delta T_{sat}^{1/0.9} \quad (56)$ where q_{cv}, q_{fc}, and q_{nc} are the total convection, forced convection, and natural convection heat transfer contributions, respectively, β is the volumetric	Mixture of ethylene-glycol and de-ionized water flowing upward in a vertical square channel with the side length of 56 mm. The range of the experimental parameters: $u \leq 2.0$ m/s $(Re_l \leq 2.9 \times 10^5)$, $q = 100\text{--}1100$ kW/m², $p = 320$ kPa, $\Delta T_{sub} = 43$ °C, $\Delta T_{sat} \leq 30$ °C.

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Table 2 (continued)

Authors	Correlations	Comments
	thermal expansion coefficient of the fluid, and $f_1(Pr_l)$ is calculated with	
	$f_1(Pr_l) = \left[1 + \left(\frac{0.492}{Pr_l} \right)^{9/16} \right]^{-16/9}$	(57)

Table 3

$q \sim \Delta T_{sat}^n$ type models of subcooled flow boiling heat transfer.

Authors	Correlations	Comments																												
McAdams et al. [22]	$q = C \Delta T_{sat}^{3.86} \quad (59)$ where the value of C ranged from 5.8 to 2.26. In some parameter ranges, the value of C decreased with decreasing the presence of air.	Degassed distilled water flowing upward in vertical annuli. Range of experimental parameters: $u = 0.3\text{--}11$ m/s ($Re_l = 3300\text{--}372,000$), $\Delta T_{sub} = 11\text{--}83$ °C, $p = 2\text{--}6.1$ bar, $D = 4.3\text{--}13.2$ m.																												
Jens and Lottes [23]	$q = \left[\frac{\Delta T_{sat}}{25} \exp\left(\frac{p}{62}\right) \right]^4 \quad (60)$ where p is in bar, and q is in MW/m².	Water data, $p < 209$ bar, $q < 12.5$ MW/m ² , $\Delta T_{sat} = 8\text{--}18$ °C.																												
Thom et al. [24]	$q = \left[\frac{\Delta T_{sat}}{22.65} \exp\left(\frac{p}{87}\right) \right]^2 \quad (61)$ where p is in bar, and q is in MW/m².																													
Kandlikar [25]	For fully developed subcooled boiling region, $q = [1058(Gh_{lg})^{-0.7}h_{sp,l}\Delta T_{sat}F_f]^{3.33} \quad (62)$ where F_f is the fluid-surface parameter depending on the fluid and surface combination, $F_f = 1$ for water, and $h_{sp,l}$ is calculated with $h_{sp,l} = Nu \frac{\lambda_l}{D} = Nu_{cp} \left(\frac{\mu_{l,b}}{\mu_{l,w}} \right)^{0.11} \frac{\lambda_l}{D} \quad (63)$ $Nu_{cp} = \begin{cases} \left[\frac{Re_l Pr_l (f/8)}{1.07 + 12.7 \left(Pr_l^{2/3} - 1 \right) (f/8)^{1/2}} \right] & 10^4 \leq Re_l \leq 5 \times 10^6 \\ \left[\frac{(Re_l - 1000) Pr_l (f/8)}{1 + 12.7 \left(Pr_l^{2/3} - 1 \right) (f/8)^{1/2}} \right] & 2300 \leq Re_l \leq 10^4 \end{cases} \quad (64)$ where the Moody friction factor is calculated with Eq. (36). For partial subcooled boiling region, $q = a + b \Delta T_{sat,ONB}^m \quad (65)$ where a, b, and m are constants obtained using heat flux at ONB and heat flux at the beginning of fully developed boiling region.	Listed below are F_f values for copper channels. For stainless steel channels, $F_f = 1$ for all fluids. <table> <tr> <th>Fluid</th> <th>F_f</th> </tr> <tr><td>Water</td><td>1.00</td></tr> <tr><td>R-11</td><td>1.30</td></tr> <tr><td>R-12</td><td>1.50</td></tr> <tr><td>R-13B1</td><td>1.31</td></tr> <tr><td>R-22</td><td>2.20</td></tr> <tr><td>R-113</td><td>1.30</td></tr> <tr><td>R-114</td><td>1.24</td></tr> <tr><td>R-134a</td><td>1.63</td></tr> <tr><td>R-152a</td><td>1.10</td></tr> <tr><td>R-31/R-132</td><td>3.30</td></tr> <tr><td>R141b</td><td>1.80</td></tr> <tr><td>R124</td><td>1.00</td></tr> <tr><td>Kerosene</td><td>0.488</td></tr> </table>	Fluid	F_f	Water	1.00	R-11	1.30	R-12	1.50	R-13B1	1.31	R-22	2.20	R-113	1.30	R-114	1.24	R-134a	1.63	R-152a	1.10	R-31/R-132	3.30	R141b	1.80	R124	1.00	Kerosene	0.488
Fluid	F_f																													
Water	1.00																													
R-11	1.30																													
R-12	1.50																													
R-13B1	1.31																													
R-22	2.20																													
R-113	1.30																													
R-114	1.24																													
R-134a	1.63																													
R-152a	1.10																													
R-31/R-132	3.30																													
R141b	1.80																													
R124	1.00																													
Kerosene	0.488																													

$$Nu_{sp} = 0.021 Re_l^{0.8} Pr_l^{0.4} \quad (21)$$

where u is the velocity of the bulk fluid, the fluid properties are valued at the local bulk fluid temperature, and the Jacob number is defined as

$$Ja = \frac{c_{pl} \Delta T_{sub}}{h_{lg}} \quad (22)$$

2.1.6. Lee–Mudawar [13] correlation

Lee and Mudawar conducted an experimental investigation of subcooled flow boiling heat transfer using HFE 7100 as working fluid and four different sizes of horizontal rectangular micro-channels. The channels were heated from bottom. Based on the experimental data, they proposed the following correlation:

$$\frac{Nu_{tp}}{Nu_{sp}} = 90.0 Bo^{0.9} Ja^{*-0.98} We^{*0.15} \beta^{-0.42} \quad (23)$$

where the boiling number Bo is defined by Eq. (11), the modified Jacob number Ja^* is given by Eq. (19), β is the ratio of channel height to channel width, and the modified Weber number is defined as

$$We^* = \frac{G^2 D}{(\rho_l - \rho_g) \sigma} \quad (24)$$

2.2. Superposition models

A superposition model considers that the heat transfer by the forced convection mechanism and that by the nucleate boiling mechanism are additive.

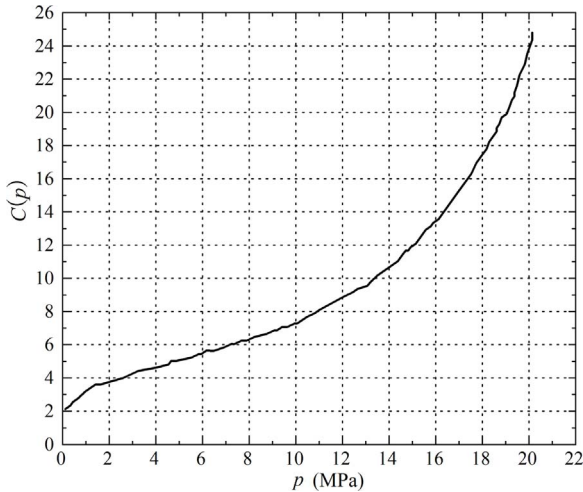
2.2.1. Rohsenow [12] correlation

Rohsenow proposed the superposition method for subcooled flow boiling heat transfer in the early 1950s as the following:

Table 4

Sources of the experimental database of subcooled flow boiling heat transfer of water.

Sources	Operation parameters: T_{sat} (°C)/ p_{sat} (MPa)/ u (m/s)/ q (MW/m ²)/ $\Delta T_{sub,in}$ (°C)/ ΔT_{sat} (°C)	Geometry: D (mm)/ L (mm)/ W (mm)/ H (mm)/Orientation/Tube shape and materials	Data points
Hua et al. [2]	99.7–120.3/0.1–0.2/0.2–1.0/0.12–0.56/4.6–25.2/25.0–58.6	17.68/300/24/14/Horizontal/Rectangular SS ^b tube	43
Hata and Masuzaki [4]	170.4/0.8/4–13.3/4.9–18.9/141.8/10.4–32.2	3/66.5/°/-/Vertical/Circular platinum tube	36
Hata and Noda [9]	158.4/0.59/6.9/6.6–17.8/125/10–32.4	9/48/°/-/Vertical/Circular SS tube	13
Hata and Masuzaki [10]	170.4/0.8/4–40/3.9–52.1/139.9/13–108.9	6/49.1/°/-/Vertical/Circular SS tube	63
Suzuki et al. [35]	99.6/0.1/0.3–0.6/3.2–9.9/30–60/22.1–209.7	1/140/1/1/Horizontal/Rectangular copper tube	62
Suzuki et al. [36]	99.6/0.1/0.5/0.5–10.2/40/4.8–92.8	① 7.37/50/5/14/② 4.94/50/3/14/; ③ 1.87/50/1/14/; ④ 14.07/50/17/12/Horizontal/Rectangular SS tubes	247
Ahmadi et al. [37,38]	98.5–173.4/0.096–0.86/0.2–1.2/0.08–0.61/4.8–59.4/3.5–18.4	13.33/400/10/20/Vertical/Rectangular SS tube	90
Okawa et al. [39]	105–106/0.12–0.13/0.09–1.5/0.067–0.497/9.2–20.8/3.6–19	20/100/°/-/Vertical/Circular glass tube	29
Phillips [40]	101–120/0.1–0.2/0.2–1.3/0.2–1.6/5–15/2–34.8	32.21/220/22.7/55.42/Vertical/Rectangular SS tube	349
Huang et al. [41]	99.7–180/0.1–1.0/0.21–0.56/0.7–6.7/5–15/17.2–65.8	10/50/°/-/Vertical/Circular copper tube	96
Weber and Johannsen [42]	102.3–180/0.11–1.0/0.1–0.11/0.2–4.6/15/17.8–49.6	10/50/°/-/Vertical/Circular copper tube	50
Cheng et al. [43]	100/0.1/0.14–0.21/0.9–2.3/14.0/30.0–50.9	12/57.2/°/-/Vertical/Circular copper tube	5
Hardik and Prabhu [44]	115.2/0.17/°/0.4–1.4/83.2–85.7/6.7–37.1	7.5–10/550–1000/°/-/Horizontal/Circular SS tube	101
Total			1184

^a Not applicable.^b Stainless steel.**Fig. 2.** Chart (based on water data) for determining $C(p)$ in Kutateladze [17] correlation.

$$q = q_{sp,l} + q_{nb} \quad (25)$$

where the single-phase forced convection component $q_{sp,l}$ and the nucleate boiling component q_{nb} are given as follows:

$$q_{sp,l} = h_{sp,l}(T_w - T_b) \quad (26)$$

$$q_{nb} = \mu_l h_{lg} \left[\frac{g(\rho_l - \rho_g)}{\sigma} \right]^{1/2} \left(\frac{c_{pl} \Delta T_{sat}}{C_{sf} h_{lg} Pr_l^n} \right)^3 \quad (27)$$

where C_{sf} and n are empirical constants depending on fluid-surface combinations, $n = 1$ for water, and $n = 1.7$ for other fluids. The C_{sf} values recommended by various researchers are listed in Table 1 [28], and $h_{sp,l}$ may be calculated with Eq. (13).

2.2.2. Butterworth's method

The extension of the Chen [29] superposition method for use in subcooled boiling region was originally proposed by Butterworth in 1970 as the following [15]:

$$q = F h_{sp,l}(T_w - T_b) + S h_{nb}(T_w - T_{sat}) \quad (28)$$

where the Reynolds number factor F is set equal to unity, $h_{sp,l}$ is calculated with Eq. (13), and the nucleate boiling heat transfer coefficient h_{nb} is obtained using the Forster and Zuber [30] correlation,

$$h_{nb} = 0.00122 \frac{\lambda_l^{0.79} c_{pl}^{0.45} \rho_l^{0.49}}{\sigma^{0.5} \mu_l^{0.29} h_{lg}^{0.24} \rho_g^{0.24}} \Delta T_{sat}^{0.25} \Delta p_{sat}^{0.75} \quad (29)$$

where Δp_{sat} is the difference of the saturation pressures corresponding to the wall temperature and the saturation temperature,

$$\Delta p_{sat} = p_{sat}(T_w) - p_{sat}(T_{sat}) \quad (30)$$

Chen [29] provided a graph (Fig. 1) for determining the suppression factor S . It was suggested that for subcooled flow boiling the effective two-phase Reynolds number Re in Fig. 1 could be approximated with Eq. (8) [15].

2.2.3. Steiner et al. [16] correlation

Steiner et al. adopted the extension of the Chen [29] correlation for use in the subcooled boiling region by Butterworth with S and F given by

$$S = \frac{1}{1 + 2.53 \times 10^{-6} (Re_l F^{1.25})^{1.17}} \quad (31a)$$

$$F = \begin{cases} 2.35(1/X_{tt} + 0.213)^{0.736} & x > 0.1 \\ 1 & x \leq 0.1 \end{cases} \quad (31b)$$

It is assumed that $F = 1$ for subcooled flow boiling since $x \leq 0.1$ in this situation. Steiner et al. verified their model with the experimental data of water.

2.2.4. Yan et al. [3] correlation

Yan et al. investigated experimentally the subcooled flow boiling heat transfer of water in a uniformly heated vertical circular tube with a 9.0 mm ID. The range of parameters covered mass fluxes of 6000, 8000, and 10,000 kg/m² s, heat fluxes of 5–12.5 MW/m², system pressures of 3, 4.2, and 5 MPa, and thermodynamic vapor quality from −0.5 to −0.03.

Based on the experimental data, a modified Chen [29] correlation was proposed. The wall heat flux q is calculated using Eq. (14), and the two-phase heat transfer coefficient h_{tp} is of the form

$$h_{tp} = F h_{sp,l} + S h_{nb} \quad (32)$$

where $F = 1$, h_{nb} is determined by Eq. (29), and S and $h_{sp,l}$ are given by

$$S = \frac{1}{1 + 2.53 \times 10^{-6} Re_l^{1.17}} \left(\frac{T_{sat}}{\Delta T_{sub}} \right)^{0.9} \quad (33)$$

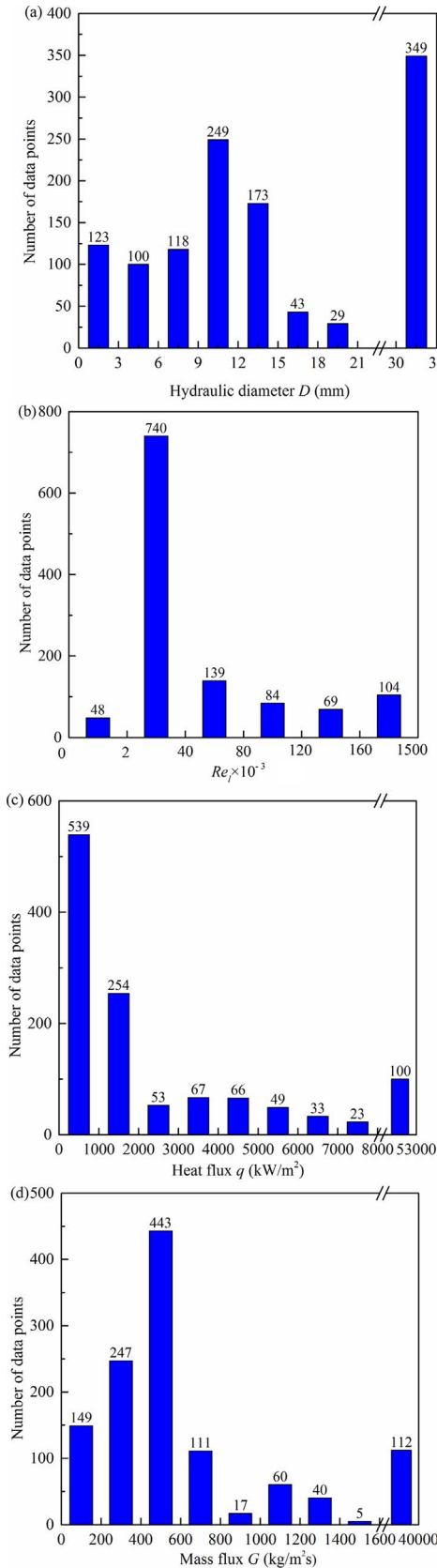


Fig. 3. Data distribution of the database related to: (a) channel size, (b) liquid Reynolds number, (c) heat flux, (d) mass flux.

Table 5

Prediction accuracy of correlations with MAD < 50% against the database.

Correlation	All data (%)		Percentile of data in given error bands (%)	
	MAD	MRD	± 10%	± 20%
Liu–Winterton [19]	32.5	2.7	22.0	39.9
Shah [5]	37.9	− 5.0	18.1	31.4
Kutateladze [17]	42.6	34.8	34.6	52.6
Baburajan et al. [11]	43.4	− 42.9	13.8	20.7
Papell [8]	43.5	− 11.3	17.1	29.8

$$h_{sp,l} = Nu \frac{\lambda_l}{D} = Nu_{cp} \left[1 + \left(\frac{D}{L} \right)^{2/3} \right] \left(\frac{Pr_l}{Pr_w} \right)^{0.01} \frac{\lambda_l}{D} \quad (34)$$

where Nu_{cp} is the Nusselt number valued at constant properties, which is calculated with the following Gnielinski [31] correlation:

$$Nu_{cp} = \frac{(f/8)(Re_l - 1000)Pr_l}{1 + 12.7(f/8)^{1/2}(Pr_l^{2/3} - 1)} \quad (35)$$

where the Moody friction factor is obtained with

$$f = (0.79 \ln Re_l - 1.64)^{-2} \quad (36)$$

2.2.5. Gungor–Winterton [14] correlation

Gungor and Winterton extended the Chen additive concept to subcooled flow boiling heat transfer and proposed

$$q = h_{sp,l}(T_w - T_b) + Sh_{nb}(T_w - T_{sat}) \quad (37)$$

where $h_{sp,l}$ is calculated with the Dittus–Boelter Eq. (13), h_{nb} is calculated by the Cooper [32] correlation, and S is determined by

$$S = 1/(1 + 1.15 \times 10^{-6} F^2 Re_l^{1.17}) \quad (38)$$

where

$$F = 1 + 2.4 \times 10^4 Bo^{1.16} \quad (39)$$

Gungor and Winterton verified their correlation with 603 subcooled flow boiling data points of water, R11, and R12, among which most are from water.

The Cooper [32] correlation is of the form

$$h_{nb} = 55 P_R^{0.12} (-0.4343 \ln P_R)^{-0.55} M^{-0.5} q^{0.67} \quad (40)$$

2.3. Asymptotic models

Asymptotic models also involve the two heat transfer mechanisms: single-phase forced convection and nucleate boiling. The general form of asymptotic models is in the power law as the following:

$$q^n = (Fq_{sp,l})^n + (Sq_{nb})^n \quad (41)$$

where n is a constant. Apparently, an asymptotic model becomes a superposition model when $n = 1$.

Table 2 lists six asymptotic models, including Hua et al. [2], Kutateladze [17], Bergles and Rohsenow [18], Liu and Winterton [19], Bjorge et al. [20], and Ramstorfer et al. [21].

2.4. $q \sim \Delta T_{sat}^n$ models

A $q \sim \Delta T_{sat}^n$ model can be reduced to

$$q = C \Delta T_{sat}^n \quad (58)$$

where coefficient C and exponent n are related to operation conditions and fluids.

The models in this category are empirical correlations generally for fully developed subcooled flow boiling heat transfer. Table 3 shows several $q \sim \Delta T_{sat}^n$ models, including those of McAdams et al. [22], Jens

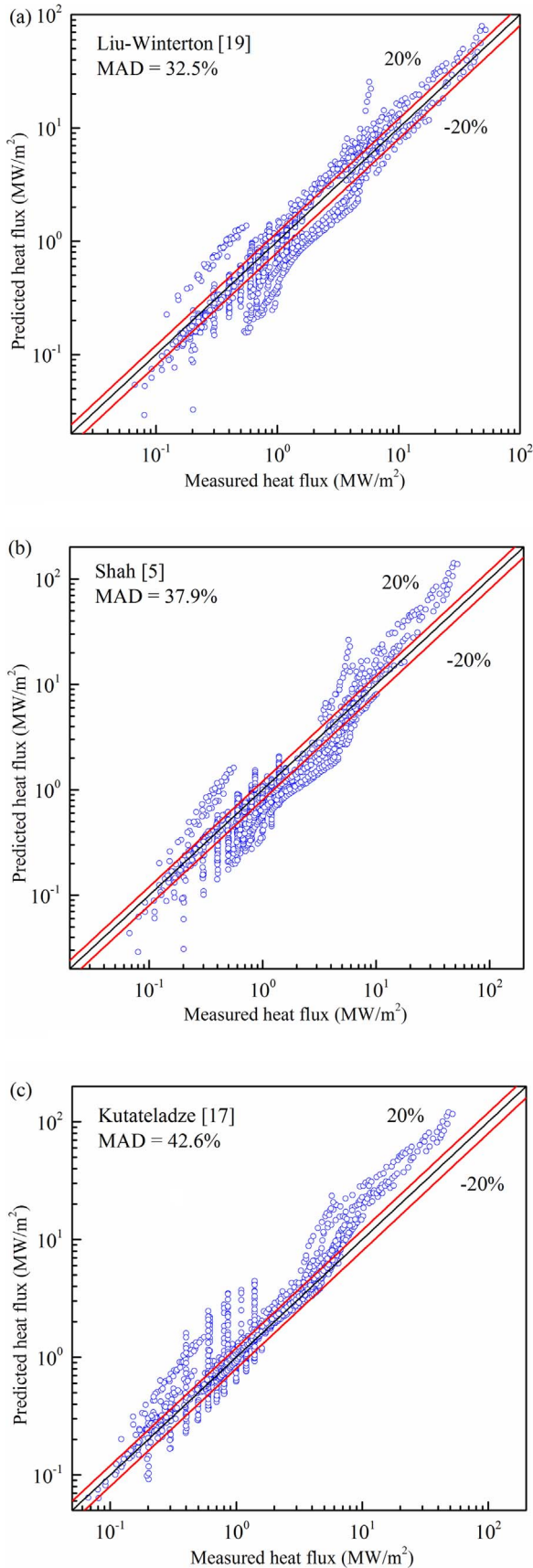


Fig. 4. Error distributions of the top correlations in prediction accuracy.

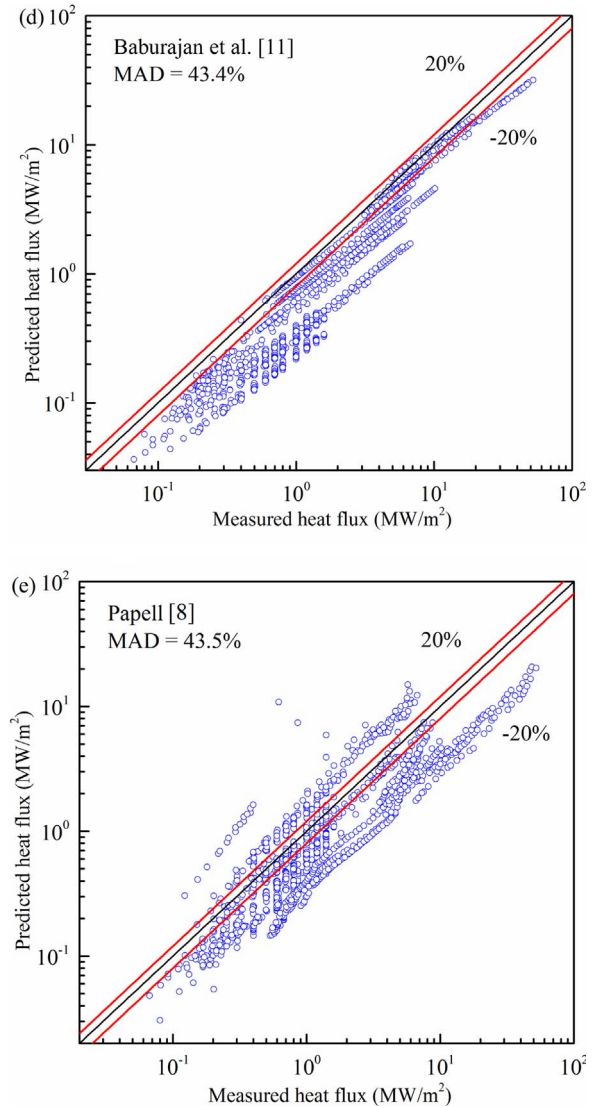


Fig. 4. (continued)

and Lottes [23], Thom et al. [24], and Kandlikar [25].

2.5. Flow pattern based models

A flow pattern based model is developed based on flow patterns. Huang and Thome [26] proposed the following model for subcooled flow boiling heat transfer:

$$Nu = \frac{h_{loc} D}{\lambda_l} \quad (66)$$

$$h_{loc} = h_{sp,loc} \left(\frac{\Delta T_{sub,loc}}{\Delta T_{sub,in}} \right) + h_{xlow} \left(1 - \frac{\Delta T_{sub,loc}}{\Delta T_{sub,in}} \right) \quad (67)$$

$$h_{sp,loc} = \frac{\lambda_l}{D} \left(\frac{4.19}{Re^{0.17} Pr^{1.43}} \right) Nu_{sp,MY} \quad (68)$$

$$h_{xlow} = 10^5 h_{3z}^{0.12} Bo^{0.39} \quad (69)$$

where $Nu_{sp,MY}$ is the single-phase Nusselt number calculated with the Muzychka and Yovanovich [33] method, and h_{3z} is the heat transfer coefficient based on the three-zone model of Thome et al. [34].

The flow pattern based method is an interesting approach and has a great potential to predict experimental data accurately. However, its accuracy depends on the accuracy of the flow pattern measurements. A

Table 6

Deviations of the top correlations in different classifications (%).

Model	$D < 3 \text{ mm}$		$D \geq 3 \text{ mm}$		Circular		Non-circular		Horizontal		Vertical	
	MAD	MRD	MAD	MRD	MAD	MRD	MAD	MRD	MAD	MRD	MAD	MRD
Liu–Winterton [19]	46.1	−1.6	30.9	3.2	32.2	21.1	32.6	−3.3	48.1	−1.1	24.8	4.6
Shah [5]	41.3	6.0	37.5	−6.3	45.7	9.9	35.3	−9.9	44.2	−4.8	34.8	−5.1
Kutateladze [17]	31.7	27.7	43.8	35.6	60.4	59.2	36.7	26.8	55.9	52.1	36.0	26.2
Baburajan et al. [11]	15.9	−15.9	46.6	−46.1	43.2	−42.6	43.5	−43.1	27.1	−26.2	51.5	−51.2
Papell [8]	31.1	−17.3	45.0	−10.6	44.4	5.6	43.2	−16.9	65.1	−41.9	32.9	3.7

Table 7

Deviations of the top correlations in each data source (%).

Model	Liu–Winterton [19]		Shah [5]		Kutateladze [17]		Baburajan et al. [11]		Papell [8]	
	MAD	MRD	MAD	MRD	MAD	MRD	MAD	MRD	MAD	MRD
Hua et al. [2]	138.5	138.5	125.6	125.6	134.2	134.2	55.4	−55.4	106.5	104.5
Hata and Masuzaki [4]	14.9	−14.9	33.8	33.8	109.9	109.9	6.8	−1.5	38.5	−38.5
Hata and Noda [9]	31.7	−31.7	6.7	−0.3	74.4	74.4	26.1	−26.1	52.3	−52.3
Hata and Masuzaki [10]	36.1	32.3	105.1	105.1	142.5	142.5	22.3	−22.3	41.4	−41.4
Suzuki et al. [35]	50.5	34.4	50.7	41.4	54.0	52.8	16.6	−16.6	29.0	−1.6
Suzuki et al. [36]	42.0	−34.1	38.6	−37.3	14.7	8.9	31.1	−31.1	43.7	−43.7
Ahmadi et al. [37,38]	15.2	−15.1	11.1	0.4	28.0	25.3	35.3	−35.3	14.5	−8.2
Okawa et al. [39]	19.6	−19.1	21.3	−11.4	9.2	3.7	50.7	−50.7	16.2	−11.5
Phillips [40]	16.5	−9.4	28.8	−27.3	14.5	−5.8	68.7	−68.7	28.6	6.2
Huang et al. [41]	33.9	33.9	29.3	−29.1	16.6	14.7	67.3	−67.3	74.8	72.6
Weber and Johannsen [42]	45.1	45.1	37.5	−34.1	33.0	33.0	49.6	−49.6	9.8	−3.0
Cheng et al. [43]	21.9	21.9	23.7	−23.7	43.8	43.8	43.3	−43.3	32.4	−32.4
Hardik and Prabhu [44]	24.5	20.0	23.4	19.3	123.2	122.5	5.4	−1.6	99.9	−99.9
Total	32.5	2.7	37.9	−5.0	42.6	34.8	43.4	−42.9	43.5	−11.3

small error in the flow pattern parameter measurements may cause a large deviation of the model. Besides, a flow pattern based model is usually complicated since it relates flow pattern information, and it is also difficult to be used since the flow pattern information needed may be unavailable in the open literature.

3. Database of subcooled flow boiling heat transfer of water

The experimental database of subcooled flow boiling heat transfer of water was compiled. It contains 1184 data points from 14 published articles, as listed in Table 4.

The ranges of the important parameters of this database are as follows:

- Diameter or hydraulic diameter $D = 1\text{--}32.2 \text{ mm}$, with 123 data having $D < 3 \text{ mm}$ and 1061 having $D \geq 3 \text{ mm}$, as illustrated in Fig. 3;
- Liquid Reynolds number $Re_l = 1016\text{--}1.5 \times 10^6$, with 96% of the data having $Re_l > 2000$ (Fig. 3);
- Heat flux $q = 0.067\text{--}52.1 \text{ MW/m}^2$, with 67% of the data having $q < 2 \text{ MW/m}^2$ (Fig. 3);
- Mass flux $G = 86\text{--}39832 \text{ kg/m}^2\text{s}$, with 80% of the data having $G < 800 \text{ kg/m}^2\text{s}$ (Fig. 3);
- Pressure $p = 0.096\text{--}1.0 \text{ MPa}$, with most of data having $p < 0.4 \text{ MPa}$;
- $\Delta T_{\text{sat}} = 2.0\text{--}209.7 \text{ }^\circ\text{C}$, $\Delta T_{\text{sub,in}} = 4.6\text{--}141.8 \text{ }^\circ\text{C}$;
- 391 data points from horizontal channels and 793 from vertical channels;
- 292 data points from circular channels and 892 from rectangular channels.

4. Evaluation of the correlations

The applicability to water of 21 models [2,3,5,8–25] is evaluated using the database. The Huang and Thome [26] correlation is not

assessed because the information in the database is not sufficient for determining h_{3z} , which is the heat transfer coefficient based on the three-zone model. The mean absolute deviation (MAD) is taken as the criterion to gauge the prediction accuracy, and the mean relative deviation (MRD) is used to check whether a correlation over-predicts or under-predicts the database.

$$\text{MAD} = \frac{1}{N} \sum_{i=1}^N \left| \frac{y(i)_{\text{pred}} - y(i)_{\text{exp}}}{y(i)_{\text{exp}}} \right| \quad (70)$$

$$\text{MRD} = \frac{1}{N} \sum_{i=1}^N \frac{y(i)_{\text{pred}} - y(i)_{\text{exp}}}{y(i)_{\text{exp}}} \quad (71)$$

where $y(i)_{\text{pred}}$ is the predicted value of point i , $y(i)_{\text{exp}}$ is the experimental value of point i , and N is the number of the total data points in the database.

4.1. Overall assessment

Table 5 lists the overall assessment results of the correlations with $\text{MAD} < 50\%$. It is seen that among the 21 correlations, only five have $\text{MAD} < 50\%$, which in order of accuracy are those of Liu–Winterton [19], Shah [5], Kutateladze [17], Baburajan et al. [11], and Papell [8], with MADs of 32.5%, 37.9%, 42.6%, 43.4%, and 43.5%, respectively. Fig. 4 illustrates the error distributions of these five correlations.

4.2. Analysis of deviations of the correlations with $\text{MAD} < 50\%$

Table 6 shows the deviations of the five top correlations in prediction accuracy for each class of the database, including $D < 3 \text{ mm}$, $D \geq 3 \text{ mm}$, circular, non-circular, horizontal, and vertical channels. $D = 3 \text{ mm}$ is a commonly-used threshold for channel size, as recommended by Kandlikar and Grande [45], which defined $D \geq 3 \text{ mm}$ as conventional channels, $200 \mu\text{m} \leq D < 3 \text{ mm}$ as mini-channels, and $10 \mu\text{m} \leq D < 200 \mu\text{m}$ as micro-channels. The Liu–Winterton [19]

correlation has the smallest MAD for the $D \geq 3$ mm (30.9%), circular (32.2%), non-circular (32.6%), and vertical (24.8%) channels, and the Baburajan et al. [11] correlation has the smallest MAD for the $D < 3$ mm (15.9%) and horizontal (27.1%) channels.

Table 7 shows the deviations of the top correlations for each data source. The results indicate that each correlation predicts the data sources very differently in accuracy. For example, the Liu–Winterton [19] correlation predicts the Hata and Masuzaki [4] data with an MAD of 14.9%, while it predicts the Hua et al. [2] data with an MAD of 138.5%. For each data source, large differences exist among the predictions of different correlations. For example, the Baburajan et al. [11] correlation predicts the Hata and Masuzaki [4] data with an MAD of 6.8%, while the Kutateladze [17] correlation predicts them with an MAD of 109.9%. These may be largely due to the application range of each correlation. In other words, most of existing correlations have very narrow application ranges.

Except for the Baburajan et al. [11] correlation, all the top five correlations predict the Hua et al. [2] data with MADs $> 100\%$. This may be due to that the Hua et al. [2] data were from a rectangular horizontal channel heated only from the bottom, indicating that a correlation developed on the basis of the data from uniformly heated channels cannot be used to predict data from non-uniformly heated channels.

5. Conclusion and discussion

- (1) The experimental database of subcooled flow boiling heat transfer of water containing 1184 data points was compiled from 14 published articles. It covers the parameter range of hydraulic diameter from 1 to 32.2 mm, heat flux from 0.067 to 52.1 MW/m², mass flux from 86 to 39,832 kg/m²s, saturation pressure from 0.096 to 1.0 MPa, inlet subcooling from 4.6 to 141.8 °C, and wall superheat from 2.0 to 209.7 °C, with 391 data points from horizontal channels and 793 from vertical channels.
- (2) All the 22 correlations, except for Huang and Thome [26] whose usage needs the flow pattern information, are assessed with the database to analyze their applicability to water. Five correlations have MADs $< 50\%$, which in order of accuracy are those of Liu–Winterton [19], Shah [5], Kutateladze [17], Baburajan et al. [11], and Papell [8], with MADs of 32.5%, 37.9%, 42.6%, 43.4%, and 43.5%, respectively. The results suggest that there is a strong need for developing an accurate correlation.
- (3) The Liu–Winterton [19] correlation has the smallest MADs for the $D \geq 3$ mm, circular, non-circular and vertical channels, with the MADs of 30.9%, 32.2%, 32.6%, and 24.8%, respectively. The Baburajan et al. [11] correlation has the smallest MADs of 15.9% and 27.1% for the $D < 3$ mm and horizontal channels, respectively.
- (4) Each of the top five correlations predicts the data sources very differently in accuracy, and for each data source, large differences exist among the predictions of different correlations, indicating that most of the correlations have very narrow application ranges, and that a new correlation should cover a wide parameter range.
- (5) The correlations of subcooled flow boiling heat transfer can be classified into five categories: enhancement-factor, superposition, asymptotic, $q \sim \Delta T_{sat}^n$, and flow pattern based types. Among the reviewed 22 correlations, the asymptotic and the enhancement-factor types each account for six, the superposition type accounts for five, the $q \sim \Delta T_{sat}^n$ type has four, and there is only one flow pattern based model.
- (6) Among the top five correlations, three belong to the enhancement-factor type, and two are asymptotic, suggesting that these two types have more potential to yield more accurate correlations.
- (7) Among the top five correlations, Shah [5] and Baburajan et al. [11] have boiling number Bo in the enhancement-factor ψ , suggesting that apart from Re and Pr , Bo is an important

dimensionless parameter for developing a correlation of subcooled flow boiling heat transfer.

- (8) The two asymptotic correlations among the top five ones, Liu–Winterton [19] and Kutateladze [17], both have power $n = 2$. In the 22 reviewed correlations, there are four methods used for determining q_{nb} or h_{nb} : the Rohsenow (Eq. (27)), the Forster and Zuber (Eq. (29)), the Kutateladze (Eq. (45)), and the Cooper (Eq. (40)) methods, among which it seems that the Cooper method is preferred. The Cooper correlation (Eq. (40)) is also a well-known nucleate boiling type correlation for saturated flow boiling heat transfer.
- (9) Among the top five correlations, there is no superposition and $q \sim \Delta T_{sat}^n$ type models. The superposition type is a special case of the asymptotic type with $n = 1$, suggesting that though the heat transfer by the forced convection mechanism and that by the nucleate boiling mechanism each may play a significant role in subcooled flow boiling heat transfer, a linear addition of these two mechanisms may be oversimplified. $q \sim \Delta T_{sat}^n$ models are generally for fully developed subcooled flow boiling heat transfer. It is understandable that a model in this type cannot perform well for a database extended to partial subcooled flow boiling. Another inherent deficiency for the $q \sim \Delta T_{sat}^n$ method is that the dimensionless approach is not used.
- (10) The flow pattern based method is an interesting approach for predicting flow boiling heat transfer. At the present, however, a flow pattern based model is usually complicated and difficult to be implemented. It deserves efforts to explore this approach.
- (11) The purpose of using the water database to evaluate the correlations is to find the better correlations suitable for water in different applications. The evaluation results cannot be construed as ranking the correlations.

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